

CHAPTER 1

ANALYSIS II SEMINAR OVERVIEW

An effective Predictive Maintenance Program (PMP) is a total program of:

- 1. DETECTION**
- 2. ANALYSIS**
- 3. CORRECTION**
- 4. VERIFICATION**

Therefore, these 4 important steps will formulate the guiding philosophy which will form the foundation for this "ANALYSIS II" seminar. Our goal will be to provide the tools an analyst needs to detect the very onset of problems within machinery included in his program. It will then provide extensive diagnostic techniques required to analyze machine problems to determine both their cause and severity. It will then put one in a position to make solid recommendations based on fact rather than "feeling", and will allow the maintenance department to schedule such corrective measures at convenient times. Finally, this seminar will provide instruction on how to verify that corrective measures did in fact correct the problem(s), and that no new problems have been introduced.

Following the completion of this course, the student should have a solid working knowledge of the proper application instrumentation and software required for both setting up and implementing an effective condition monitoring program, as well as significantly enhance his knowledge on how to effectively troubleshoot mechanical and electrical problems within machinery using vibration analysis and related nondestructive technologies. Following below are brief introductions for each of the chapters which will be covered in this seminar text:

- **CHAPTER 2 - "BRIEF REVIEW OF 'ANALYSIS I' SEMINAR TOPICS":** This chapter will review some of the more important topics which are covered in the "ANALYSIS I" seminar. Initially, it will review the fundamentals of vibration analysis and how it can be used to evaluate machine condition. Next, it will provide a generic overview of the various types of vibration instruments available today, including both their strengths and their weaknesses. This includes a comprehensive table which summarizes a great number of capabilities which are possible with vibration analysis, and clearly identifies which of the instrument types (not specific vendors) can perform which of the tabulated tasks. Finally, the chapter closes by reviewing each of the major vibration transducers available today and gives important instruction on how to properly select the right transducer for the particular job or test to be performed.
- **CHAPTER 3 - "PRINCIPLES OF DIGITAL DATA ACQUISITION AND FFT PROCESSING FOR RELIABLE SPECTRAL ANALYSIS":** This comprehensive chapter provides the analyst with important information on how his vibration analyzer acquires and processes the data which is so critical to the success of his program. Often, seminars simply assume the analyst understands these principles and never provide any real background on just how the instruments function, what effects it might make on their data if they do not understand this information, and how they might best optimize their instrument and supporting software to acquire the data they need to properly evaluate the machines or structures in question. More specifically, this chapter will examine FFT properties, sampling and digitizing of data, aliasing, window selection, types of averaging, overlap processing, the importance of bandwidth in ensuring all frequencies from all sources are displayed, the effect of dynamic range on frequency and amplitude display, and a discussion on the distinct difference between digital and analog acquisition of the overall vibration level.

● **CHAPTER 4 - "INTRODUCTION TO NATURAL FREQUENCY TESTING & INSTRUMENTATION"**

This chapter will introduce the analyst to the difference between the terms "natural frequency", "resonance" and "critical speed". It will then provide invaluable information on how to perform a variety of natural frequency tests, including both the instrumentation and accessories required to accurately perform these tests. Test methods taught will include impulse natural frequency tests, coastdown and runup tests; and how to both acquire and interpret Bode' and polar plots which confirm the presence of natural frequencies (some of which may be found to be resonances of the machinery being evaluated). In these cases, introductory information will be provided on how to go about correcting such resonance problems.

● **CHAPTER 5- "ENHANCED VIBRATION DIAGNOSTICS USING PHASE ANALYSIS AND CASCADE DIAGRAMS"**

Describes the various instruments and transducers required to measure phase as well as those to generate cascade diagrams (sometimes called "waterfall plots" or "spectral maps"). This chapter points out that phase is the "third leg of the triangle" which describes machine and structural vibration response. That is, these "three legs of the triangle" include amplitude (how much vibration), frequency (how many cycles of vibration per unit of time) and phase (which describes the vibration at one location relative to the vibrating motion at another location). This chapter points out that by taking phase measurements in the horizontal, vertical and axial directions on each bearing housing, one can determine whether a problem showing high vibration at 1X RPM is unbalance, misalignment, soft foot, bent shaft, eccentric rotor, loose hold-down bolts, resonance, cocked bearing, a combination of these problems, or several other potential problems (all of which can generate vibration spectra which "appear" to be identical). This chapter then takes a close look at the value and optimum utilization of cascade diagrams pointing out how they give the analyst a unique view of how the vibration response changes either over a short period of time (for example, during either a runup or coastdown), or over a rather long period of time (for example, from one month to the next during PMP surveys). Instruction is provided on how to use this information to help diagnose a number of problems including rotor rub, resonance, oil whirl, oil whip, etc.

● **CHAPTER 6 - "CONCENTRATED VIBRATION SIGNATURE ANALYSIS TO DETECT A SERIES OF MECHANICAL AND ELECTRICAL PROBLEMS"**

Chapter 6 probably forms the "centerpiece" of this seminar and has been widely acclaimed by attendees over the years as the key which has helped them significantly elevate the effectiveness of their programs. This chapter will introduce Technical Associates' world renown "**Illustrated Vibration Diagnostic Wall Chart**" which will review theory on how to detect some 44 machine problems, including those from both mechanical and electrical problem sources. This chapter begins with a review of the less complex problems which are covered in the "ANALYSIS I" seminar; and then provides in-depth instruction on how to detect rotor rub, journal bearing, gear, electrical, beat vibration, resonance and rolling element bearing problems (including a series of "failure scenarios" which have been identified through the years to track the condition of rolling element bearings).

● **CHAPTER 7 - "PROVEN METHOD FOR SPECIFYING BOTH SIX SPECTRAL ALARM BANDS AS WELL AS NARROWBAND ALARM ENVELOPES USING TODAY'S PREDICTIVE MAINTENANCE SOFTWARE SYSTEMS"**

After covering how to detect the whole series of mechanical and electrical problems, as well as revealing where they will appear in a vibration spectrum in Chapter 6, Chapter 7 next shows one how to properly specify both spectral band as well as narrowband envelope alarms, not only to detect the presence of such problems, but also to give the analyst plenty of time to react and take the required corrective measures before potential catastrophic failures. These documented methods (which were developed over the past 15 years by implementing a series of predictive maintenance programs on a broad range of machinery) have received wide acclaim and, like the signature analysis theory taught in Chapter 6, have often received much credit for greatly enhancing the effectiveness of condition monitoring programs by allowing the analysts to concentrate their efforts on the machines truly in need of attention.

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- **CHAPTER 8 - "INTRODUCTION TO LISSAJOUS ORBIT ANALYSIS"**: Introduces the analyst to Lissajous orbit plots which show the actual path the shaft itself follows inside the bearing. It points out that Lissajous patterns can be used to study the shaft dynamic behavior, measure the relative phase angle between motions at different points on the structure, and to detect the presence of machine faults such as mechanical runout, eccentricity, misalignment, rotor rub, gear and bearing faults. This chapter provides this information in practical terms and provides real-world examples of how Lissajous pattern recognition has been so successful in detecting numerous problems which might otherwise have gone undetected had this technique not been employed.
 - **CHAPTER 9 - "INTRODUCTION TO HIGH FREQUENCY DEMODULATED AND ENVELOPED SPECTRA"**: Describes how these tools which are now available on many of today's programmable data collectors can be used to provide an early warning of impending problems with rolling element bearings, gears, cavitation, lubrication, electrical faults, etc. Although these techniques have been around for some years, only recently since approximately 1990 have they began to appear within many of today's data collectors. The problem is that so few analysts yet understand what high frequency enveloping technology is, much less how it works. Therefore, the expressed purpose of this chapter is to initially provide the analyst with a fundamental understanding of how this data is acquired, how it is processed and how to interpret the resulting spectra once they have been generated. Then, if this tool is used along with vibration spectral analysis, they provide a powerful set of diagnostic tools which can not only detect problems at early stages, but also can track their deterioration and allow their correction before the damage to the machine is extensive.
 - **CHAPTER 10 - "INTRODUCTION TO VIBRATION ISOLATION MECHANISMS"**: Introduces the analyst to a variety of vibration isolation mechanisms and points out the distinct difference between the terms "vibration isolation" and "vibration damping" which unfortunately are often used by many to mean the same thing. This chapter, along with Chapter 11, points out the distinct difference between the two terms and describes how each of these two vibration treatment methods function on a very practical level. This chapter not only discusses the theory and provides illustrated examples of some of the more popular isolators available, but also gives good "rules of thumb" on how to specify proper isolation treatments (in order to avoid amplifying vibration rather than isolating it). Likewise, it includes an invaluable table comparing many of the isolator types and showing what frequencies they will and will not isolate. This chapter also includes a real-world example of how isolation was employed to prevent vibration originating from a two-stage reciprocating air compressor from transmitting into an electron microscope directly above it on the second floor even though the compressor itself was installed only 21 inches away from a load bearing wall common to the building structure of both the compressor room and microscope lab.
 - **CHAPTER 11 - "INTRODUCTION TO VIBRATION DAMPING TREATMENTS"**: Introduces the analyst to the theory of vibration damping on a practical level. This chapter points out this parameter (damping) is probably the most misunderstood of any of the three major parameters effecting vibration response of a machine or structure (stiffness, mass and damping). It provides a good definition and description of the theory of damping in everyday, practical terms rather than the usual, highly technical jargon normally associated with discussions on this topic. Likewise, it also provides information on some of the more popular damping treatments; and, importantly, points out when damping treatments should be used, as well as when installation of damping materials is likely to be a waste of time and funding.

● **CHAPTER 12 - "REAL-WORLD CASE HISTORIES OF VIBRATION DIAGNOSTICS**

CONDUCTED ON VARIOUS MACHINE TYPES": Offers an array of actual case histories which have been performed in order to give the student a taste of how such problems were solved on actual machines using the tools taught in the seminar. This section includes over 250 pages of such case histories showing how such problems as rotor rub, gear, electrical, resonance and rolling element bearing wear were detected, and subsequently corrected without catastrophic failure. Impressive "before" and "after" frequency spectra are included to show the effect of properly taking the recommended corrective actions on the machine, and thereby prolonging the life of the equipment by reducing these vibration amplitudes.